

Machine learning for healthcare: landscape

INF-GRAM — Machine Learning for Healthcare

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Session roadmap



What this first lecture covers

Today we will answer four simple questions:

1. **Why** is machine learning useful in healthcare?
2. **What** kinds of data do we work with?
3. **Which** broad families of methods should we distinguish?
4. **How** should a real machine-learning project be organized?

Learning objectives

By the end of the lecture, you should be able to:

- distinguish supervised and unsupervised learning;
- describe the main stages of a machine-learning workflow;
- identify the main challenges of healthcare data;
- explain why evaluation and deployment matter as much as model fitting.

Why healthcare is a natural application area

Why people are interested in ML for health

Healthcare produces large amounts of information, often with high stakes:

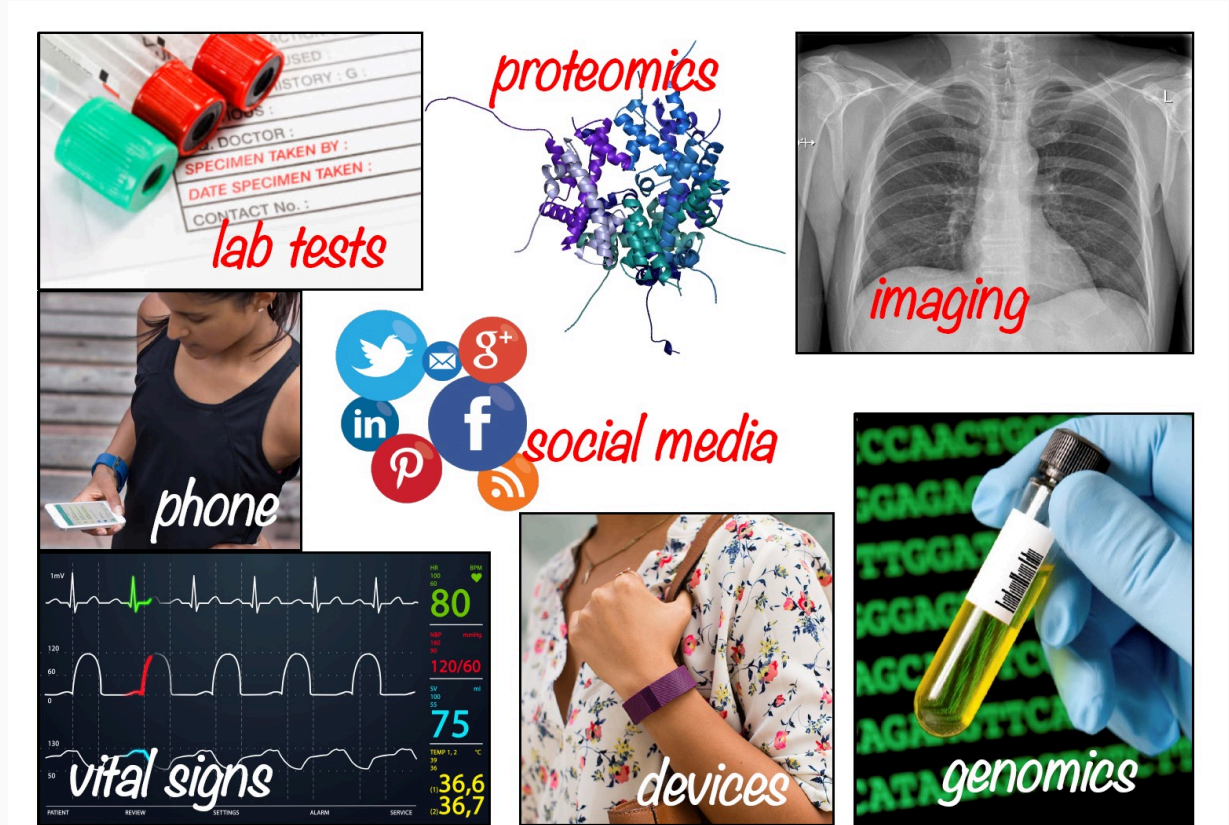
- diagnosis and prognosis;
- treatment choice;
- risk stratification;
- workflow optimization;
- population-level monitoring.

This makes healthcare attractive for data-driven methods, but also demanding.

A wide variety of data sources

Typical sources include:

- electronic health records;
- lab measurements;
- medical imaging;
- clinical notes;
- physiological signals;
- data produced by patients themselves.



More data does not automatically mean better learning

In healthcare, data are often:

- incomplete;
- noisy;
- collected for care rather than research;
- heterogeneous across hospitals;
- affected by coding habits and organizational constraints.

So the first challenge is usually **data quality**, not algorithm choice.

From raw information to usable data



Why structure matters

Before learning anything useful, many healthcare variables must be:

- standardized;
- coded;
- documented;
- linked to a clear clinical meaning.

For example, a database may store:

- diagnosis codes;
- procedure codes;
- medication categories;
- laboratory vocabularies.

This makes large-scale analyses possible.

ICD-9 codes 290–319: mental disorders

ICD-9 codes 320–359: diseases of the nervous system

ICD-9 codes 360–389: diseases of the sense organs

ICD-9 codes 390–459: diseases of the circulatory system

ICD-9 codes 460–519: diseases of the respiratory system

ICD-9 codes 520–579: diseases of the digestive system

ICD-9 codes 580–629: diseases of the genitourinary system

ICD-9 codes 630–679: complications of pregnancy, childbirth,

But structure is not the same as truth

Even a nicely coded dataset can still contain:

- measurement errors;
- missing values;
- selection bias;
- delayed recording;
- labels that do not correspond exactly to the question of interest.

This is why domain knowledge remains essential.

What do we mean by machine learning?

A practical definition

A machine-learning model learns a rule from data in order to:

- **predict** an outcome;
- **classify** an observation;
- **estimate** a quantity;
- **discover** hidden structure.

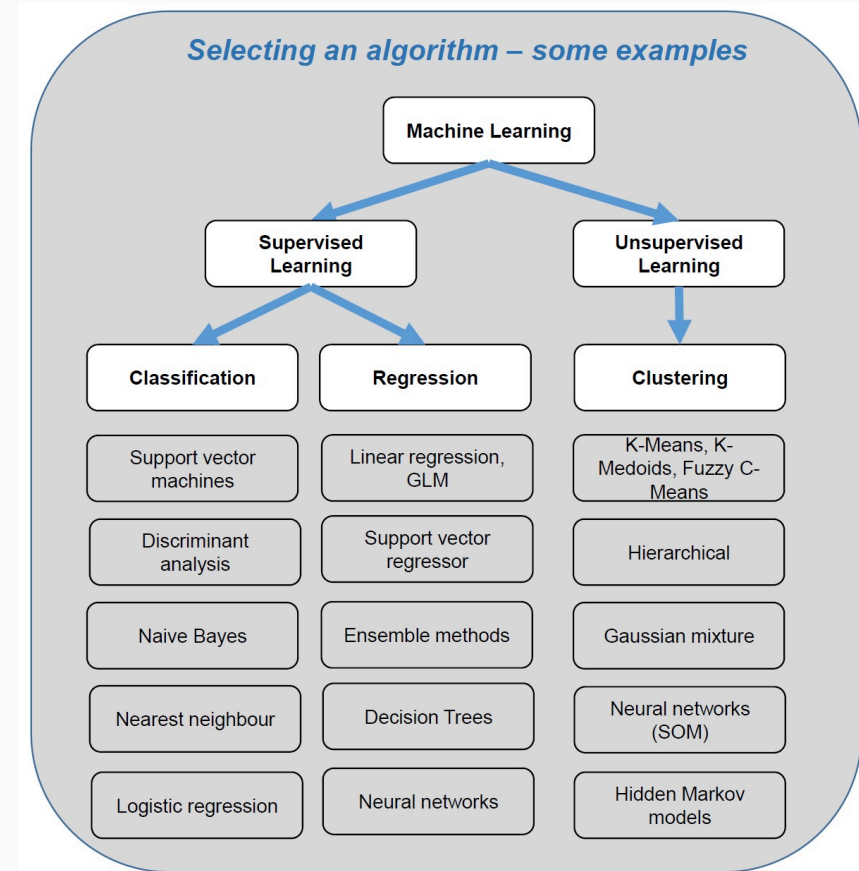
A useful way to think is: **task + data + evaluation**.

Main families of methods

The two big families we will use in this course are:

- **supervised learning**
 - classification
 - regression
- **unsupervised learning**
 - clustering
 - dimensionality reduction

The distinction depends on whether we have a target variable.



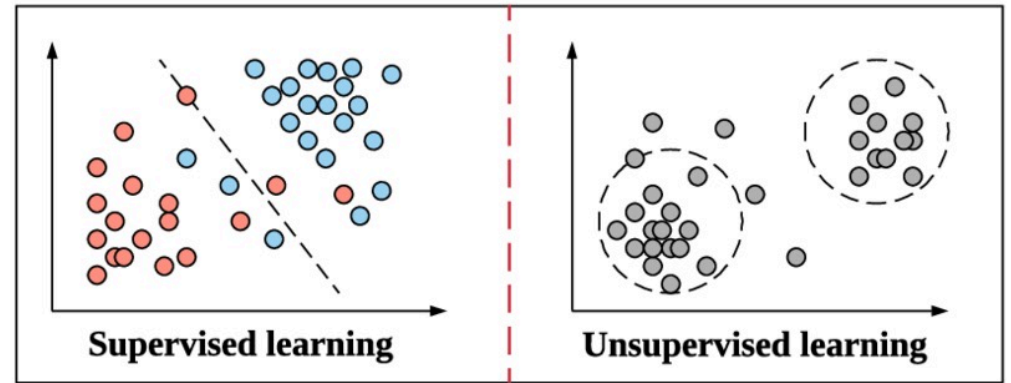
Supervised vs unsupervised learning

Supervised learning

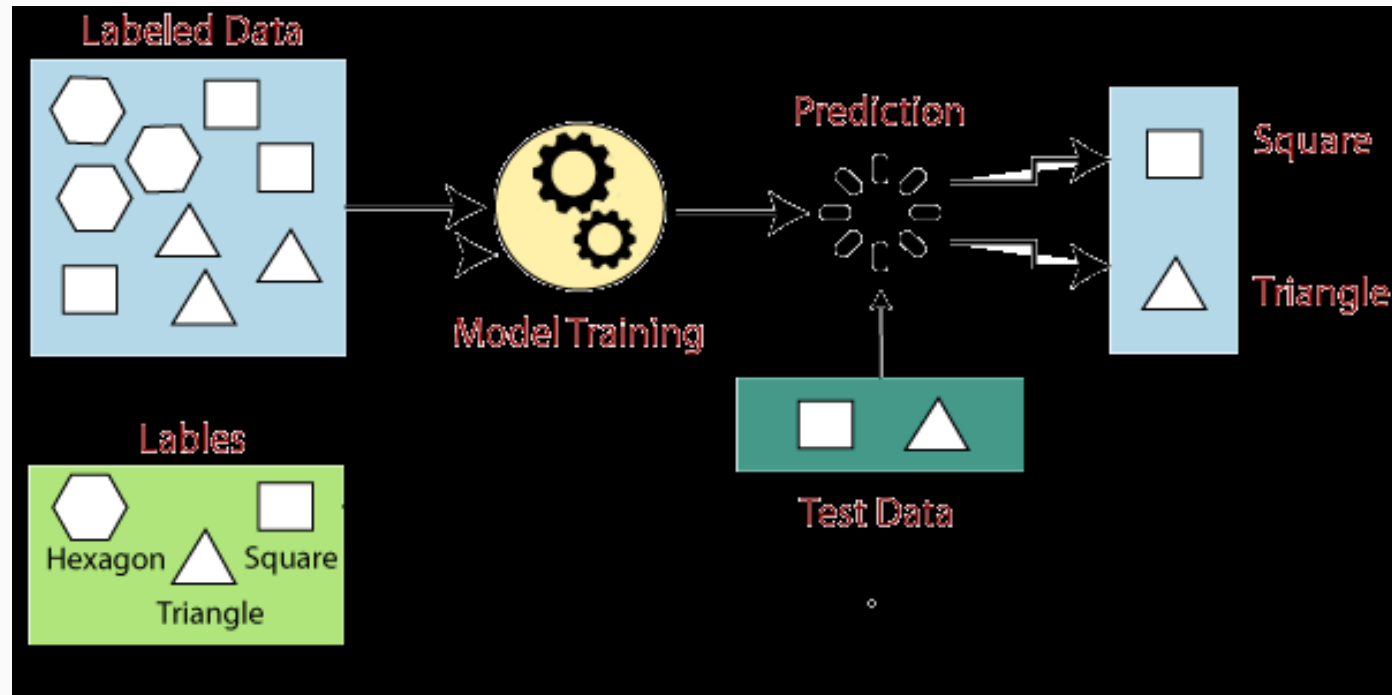
- inputs (X) and known outcomes (y);
- goal: learn to predict the outcome;
- examples: readmission, mortality, diagnosis.

Unsupervised learning

- no target variable;
- goal: explore structure in the data;
- examples: patient phenotypes, clusters, profiles.



Example of a supervised task



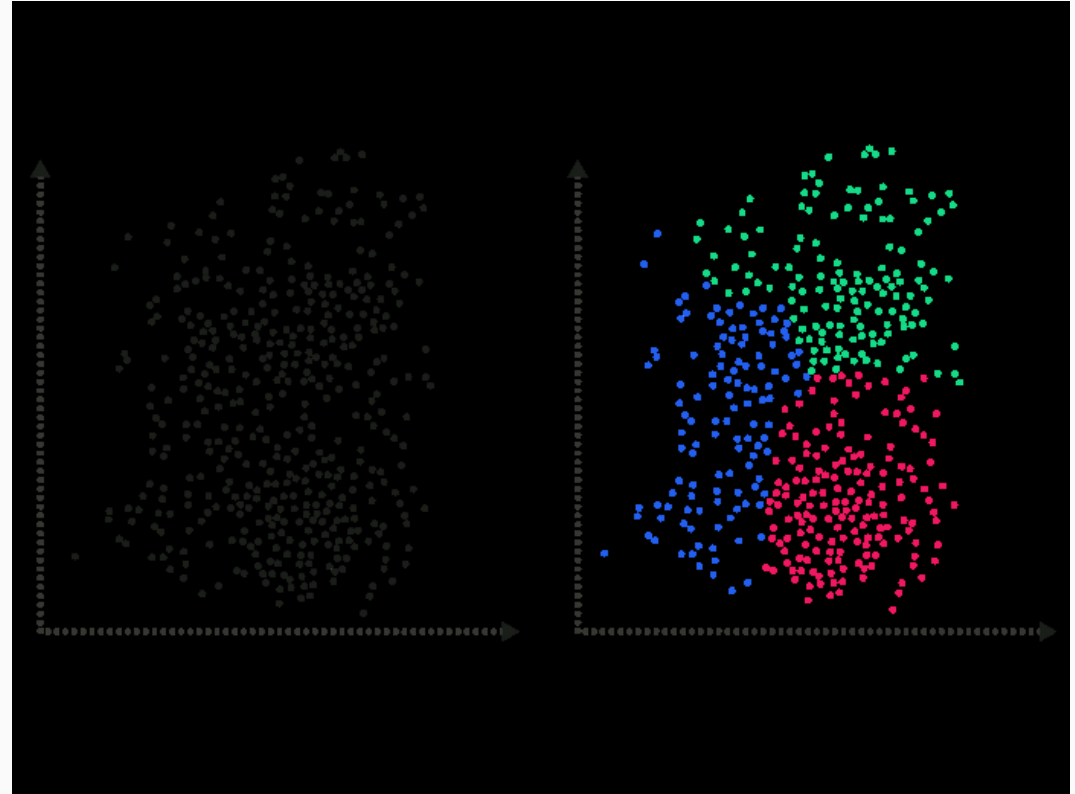
A labelled training set lets us learn a rule that we can apply to a new patient.

Example of an unsupervised task

Clustering can help us:

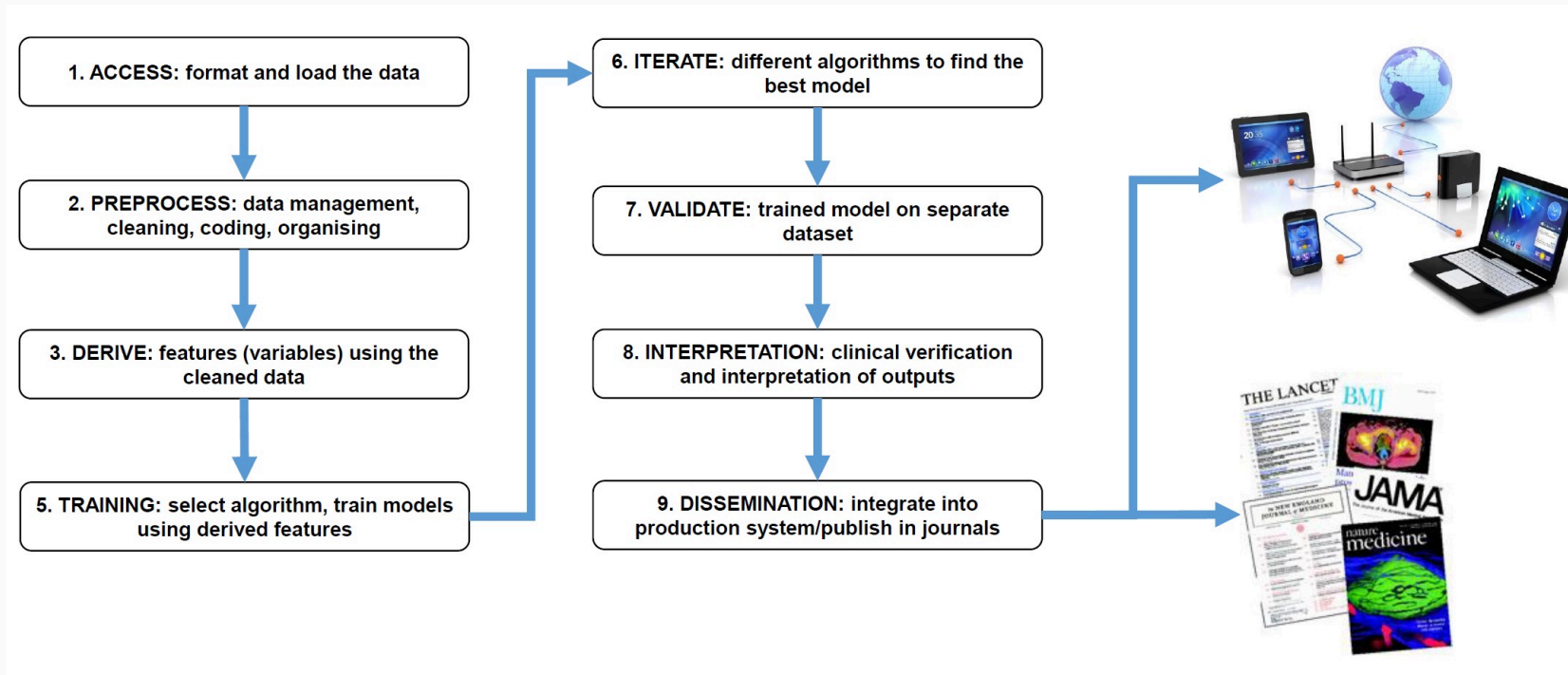
- segment patients into groups;
- explore heterogeneity;
- generate hypotheses;
- produce simpler summaries of complex data.

But the discovered groups are not automatically clinically meaningful.



A machine-learning project is a workflow

A realistic workflow



The key message

A successful project is **not** only about training a model.

It also requires:

1. a precise clinical question;
2. a relevant target population;
3. careful data preparation;
4. appropriate validation;
5. interpretation and communication.

Common pitfalls in healthcare ML

Typical failure modes include:

- data leakage;
- overfitting on small samples;
- poor external validity;
- unrealistic performance claims;
- lack of interpretability or clinical usefulness.

A technically sophisticated model can still answer the **wrong question**.

Take-home messages

What to remember from this lecture

- Healthcare data are rich, varied and difficult to use well.
- Supervised learning predicts an outcome; unsupervised learning discovers structure.
- The success of a project depends on the full workflow, not only on the algorithm.
- In healthcare, the most important questions are often about **data quality, validation and clinical relevance**.